QP Code: 25900262

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Elective (DSE)-MG3DSEBOTA03-FUNDAMENTALS OF PLANT BIOTECHNOLOGY & INDUSTRIAL BOTANY

(2024 ADMISSION ONWARDS)

Duration 2 hrs **Max mark 70**

Part A One or two Sentences

Answer **any 10** questions. Each question carries **2 marks**

1 Recall any two biosafety concerns of GMOs

A: Risks to human health, toxicity, allergens**.(***any two points)*

2 What is cryopreservation?

A: Cryopreservation is the preservation of living cells and tissues in liquid nitrogen (LN) at − 196 °C for a longer period without compromising biological functionality.

3. Write a short note on plant biotechnology

A:Plant biotechnology is the use of scientific techniques to modify and improve plants for specific purposes, such as agriculture, industry, and medicine

4. What is an explant?

A: An explant is a small piece of tissue or a few cells taken from the parent plant and placed on a sterile, nutrient-rich medium in vitro growth to form large number of clonal plants.

5. How does polymeric encapsulation differ from alginate encapsulation in synthetic seed production?

Alginate encapsulation	**Polymeric encapsulation**
Material- sodium alginate	Material includes any type of polymer like gelatine, polyoxyethylene glycol, etc.
A simple, fast, and low-toxicity ionotropic gelation process	Synthetic polymers may require more complex processes and may involve desiccation for long-term storage.
Hydrated, gel-like bead that is biocompatible, non-toxic, and porous.	The properties of beads are highly dependent on the specific polymer used

(or any two valid points)

6. Name two causes of somaclonal variation.

A: Chromosome abnormalities, abiotic stress, abnormal cell division, mutations in gene, epigenetic changes, transposable elements, genetic changes in cytoplasm, age of culture, etc. *(Any two valid points)*

7. **What is the primary purpose of a plant nursery?**

A: **:** Mass propagation plants, grow plant varieties for customer demand, use controlled environment conditions for plant growth, conservation of plant varieties for the enrichment of biodiversity, plant science research. *(any one point)*

8. **How do you control temperature and humidity in a nursery?**

A: Proper ventilation, proper heating / cooling systems, digital thermometer to measure the temperature, humidifier and dehumidifier. *(any two methods)*

9. Define Floriculture.

A: Floriculture is the professional cultivation of flowers and decorative plants. Floriculture is the cultivation, processing, marketing, and arrangement of flowers and decorative plants for a variety of purposes, including decoration, commerce, and medicine.

10. What crops are suitable for organic farming?

A: Any two economically important crops

11. Describe the nutritional benefits of straw mushrooms

A: *Volvariella volvacea* consist of moisture, fiber (chitin), vitamins (thiamine, biotin, riboflavin and high quantity of vitamin C), protein, carbohydrates, fats, essential amino acids (glycine, arginine, alanine and serine, etc.), important minerals (sodium, potassium and phosphorus), unsaturated fatty acids, and also have low calorific value

12. Write short notes on microgreens.

A: Microgreens are young and tender edible greens produced by sprouting the seeds of a variety of vegetable species and herbaceous plants

Part B- Short Answer

Answer any 8 questions. Each question carries 5 marks

13. Genetic engineering is influential in developing novel modifications in plants. Explain

A: Pest and Disease Resistance**,** Herbicide Tolerance, Nutritional Enhancement, Abiotic Stress Tolerance. (*Any three valid points with brief explanation)*

14. Describe the importance of differentiation, dedifferentiation and redifferentiation in plant tissue culture

A: Differentiation, dedifferentiation, and redifferentiation are fundamental concepts in plant tissue culture, central to the ability to regenerate a whole plant from a single cell or small piece of tissue.

15. Enumerate the applications of Agricultural Biotechnology.

A: Increased Yield and Quality, Biofortification, Biopharmaceuticals and Edible Vaccines, Herbicide Tolerance. *(Any four points)*

16. **Explain the significance of protoplast fusion in the creation of somatic hybrids and cybrids.**

A: Protoplast fusion is the biotechnological process involving the aseptic mixing up of protoplasts from two different plants. This process is used to create new varieties with desirable traits, such as disease resistance, by either combining the entire genomes to form a somatic hybrid or fusing nuclei

from one parent with cytoplasm from both to create a cybrid. It is helpful in overcoming sexual barriers, crop improvement and overcoming sexual barriers, crop improvement, genetic research, novel plant varieties and cytoplasmic inheritance. (*any three valid points*)

17**. Analyse the limitations of synthetic seed technology in field-level propagation.**

A: Low survival rate, lesser percentage into next developmental stage, lack of dormancy, inability to tolerate stress, unorganised germination, high initial cost, scalability problems, incompatability in mechanised farming, toxicity, nutrient availability of coating materials. (*any five valid points*)

18. What is the importance of callus initiation in micropropagation?

A: Callus culture is an in vitro technique used in plant tissue culture to grow and maintain an unorganized, undifferentiated mass of plant cells called a callus. It helps in plant regeneration, mass propagation, genetic modification, conservation of rare endangered plants, secondary metabolite production, single cell cultures, cultivation of plants independent of seasonal limitations, it is a tool for studying plant development. (*any three valid points*)

19. Why are seed industries important for agriculture?

A: Economic Benefits for Farmers, Food and Nutritional Security, Increased Crop Yields, Risk Mitigation for Farmers**.**(Any Four points with brief explanation can give full credit)

20. Dry flowers are good prospects for revenue generation, Validate the statement with examples.

A: Floriculture and products from dry flowers are good sources of revenue generation it is supported by a growing global market with high profits with a longevity and versatility than fresh flowers. Dry flowers are used to generate numerous products with better export opportunities and creates a new opportunity for employability. Products from dry flowers include: home decors, potpourri, handicrafts, dry flower themed events, beverages, essential oil and perfume extraction, cosmetics and skin care products, resin art products, etc (*any 5 products from dry flowers may be mentioned)*

21. Describe the role of seed processing for seed industry.

A: Enhancing Seed Quality and Purity, Improving Germination and Vigor, Ensuring Safe Storage and Increased Longevity, Standardization and Uniformity, Disease and Pest Protection (Seed Treatment), Increasing Market Value, Efficient Handling and Logistics, info traceability *(Any 4 points with brief explanation can give full credit)*

22. Describe a method for controlling pests and diseases in mushroom production.

A: Sanitation and hygiene are the most important, avoid using damaged and old straw for bed preparation, remove and destroy the infected beds immediately, proper sterilization of straw and bed. **(***Any one method)*

Part C- Essay

Answer any 1 question. Each question carries 10 marks

23. **Discuss the techniques and significance of somatic hybridization and cybridization.**

A: Somatic hybridization and cybridization are biotechnological techniques that enable the combination of genetic material from different plant species, overcoming barriers of sexual incompatibility.

Somatic Hybridization
This method fuses protoplasts—plant cells without cell walls—from two species to form a hybrid containing both nuclear and cytoplasmic genomes.
Technique: Protoplasts are first isolated by enzymatically removing cell walls using cellulase and pectinase. The protoplasts are then cultured in a nutrient medium and fused using chemical agents like polyethylene glycol (PEG) or electrical pulses. Hybrid cells are selected through metabolic markers or fluorescence and regenerated into whole plants.
Significance: Somatic hybridization enables the creation of novel genetic combinations which are not possible through sexual crosses, including interspecific or intergeneric hybrids. It can produce fertile plants from sterile parents and introduce valuable cytoplasmic traits like disease resistance or stress tolerance. Applications include developing disease-resistant *Citrus* varieties and improving fruit quality in persimmons.

Cybridization
Cybridization, or cytoplasmic hybridization, involves the fusion of a whole protoplast from one parent with an **enucleated** protoplast (cytoplast) from another. The resulting cybrid carries the nuclear genome of one species and the cytoplasm of both.
Technique: One parent's cells are enucleated, and these cytoplasts are fused with intact donor protoplasts. The resulting cells are cultured and regenerated into plants.
Significance: Cybridization allows selective transfer of cytoplasmic traits—such as cytoplasmic male sterility, herbicide tolerance, or disease resistance—without altering the nuclear genome. It is particularly useful in breeding programs to introduce desirable cytoplasmic characteristics while retaining the superior nuclear background of a cultivar.

24. What is organic farming? Discuss the benefits and challenges of organic farming.

A: Organic farming is a sustainable agricultural system that avoids synthetic inputs like pesticides, fertilizers, and GMOs, and instead emphasizes natural processes and materials to build soil health and fertility.

Benefits of Organic Farming:

- Economical Farming
- Environment Friendly
- Sustains Soil Fertility
- Provides Healthy Food
- Inexpensive technique
- Generates Income through Exports

Challenges before Organic Farming

- Less Popular
- Lack of Infrastructure and Marketing Facilities
- Low Yield
- Shorter Food Life
- Limited Choice of Crops

(Definition 2marks + Benefits 4 marks + Challenge 4marks)
